DISCRETE MATH · ELEMENTARY NUMBER THEORY AND METHODS OF PROOF

# Euclidean Algorithm Explorer

Explore & Reflect

Name _____ Date _____ Period _____

Open **Euclidean Algorithm Explorer** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

## 1 Predict before you touch the tool

1. Without the tool, guess gcd(84, 36). How did you guess?

2. If a = bq + r, why should a and b have the same common divisors as b and r?

## 2 Explore use the tool

1. a = 252, b = 198. Step through the cuts. Record each equation a = bq + r.

2. What is the side of the smallest squares? How does it relate to the last equation?

3. Find a pair under 100 that needs at least 6 steps. What kind of numbers are they?

4. Use the Quotient and remainder tab with n = −23, d = 7. Record q and r.

## 3 Record use the tool and fill in the table

**GCD, LCM and a linear combination**

Use Show all steps. Check that gcd · lcm = ab.

| a | b | Steps | gcd(a, b) | lcm(a, b) | gcd = s·a + t·b |
|---|---|---|---|---|---|
| 84 | 36 | | | | |
| 91 | 120 | | | | |
| 72 | 60 | | | | |
| 89 | 55 | | | | |

## 4 Explain what did you discover?

1. gcd(a, b) = gcd(b, _____), where r is the remainder when a is divided by b.

2. The GCD is the last _____ remainder.

3. $\gcd(a, b) \cdot \operatorname{lcm}(a, b) =$ _____.

4. For any integer $n$ and $d > 0$, $n = dq + r$ with _____ $\le r <$ _____.

5. Why must the Euclidean algorithm stop? Use the remainders in your answer.

## 5 Apply a new situation, no tool

1. Use the Euclidean algorithm to find $\gcd(1071, 462)$.

2. Find $\operatorname{lcm}(1071, 462)$.

3. Write $\gcd(30, 21)$ as a combination $30s + 21t$.

4. Find $-40 \operatorname{div} 6$ and $-40 \bmod 6$.